

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Practice Program

Date: 30/07/2026

Week No.: 4

Program S. No. 7

Title	Feature Engineering and Correlation Analysis Using Pandas			
Problem Statement	Create a DataFrame with columns Feature1, Feature2, and Target containing predefined sample data stored in a Python dictionary. Create a new feature that is the logarithm of Feature1, bin Feature2 into three categories (Low, Medium, and High), and calculate the correlation matrix of the DataFrame. Finally, create a scatter plot of Feature1 versus Feature2 distinguished by Target and interpret the visible patterns.			
Objectives	• To create a Pandas DataFrame using a predefined Python dictionary. • To understand feature engineering using logarithmic transformation. • To transform Feature1 using the natural logarithm function. • To divide Feature2 into Low, Medium, and High categories. • To calculate and interpret correlations among numerical variables. • To visualize the relationship between Feature1 and Feature2 using a scatter plot. • To distinguish Target classes using a black-and-white plotting format. • To interpret visible trends, class overlap, and feature relationships.			
Dataset URL Link / File	Predefined dictionary dataset used directly inside the Python program:			
	Record	Feature1	Feature2	Target
	1	5	10	0
	2	8	18	0
	3	12	15	1
	4	15	25	0
	5	18	30	1
	6	22	28	1
	7	25	35	0
	8	30	40	1
	9	35	38	1
	10	40	45	0
	11	45	50	1
	12	50	55	1
	13	55	60	0
	14	60	65	1
	15	70	75	1
No external CSV file, URL, database, or online dataset is required.				
Functions Details	Sl. No.	Function	Package	Description of function
	1	pd.DataFrame()	Pandas	Creates a DataFrame from the predefined dictionary data.
	2	np.log()	NumPy	Calculates the natural logarithm of each Feature1 value.
	3	pd.cut()	Pandas	Bins Feature2 into Low, Medium, and High categories.

Student Reg. No: 26MML0031 Name: Makada Hasnain

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	4	corr()	Pandas	Calculates pairwise correlations between numerical columns.
	5	value_counts()	Pandas	Counts the number of records in each Feature2 category.
	6	sort_index()	Pandas	Displays category counts in their defined category order.
	7	round()	Python/ Pandas	Rounds logarithmic and correlation values for readable output.
	8	plt.figure()	Matplotlib	Creates a plotting area with the specified dimensions.
	9	plt.scatter()	Matplotlib	Creates the Feature1 versus Feature2 scatter plot.
	10	plt.xlabel()/plt.ylabel()	Matplotlib	Adds descriptive labels to the plot axes.
	11	plt.legend()	Matplotlib	Identifies the symbols used for each Target class.
	12	plt.show()	Matplotlib	Displays the completed scatter plot on the screen.
Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	Step 1: Import Pandas, NumPy, and Matplotlib. Step 2: Define Feature1, Feature2, and Target values inside a Python dictionary. Step 3: Convert the dictionary into a DataFrame using pd.DataFrame(). Step 4: Display the original DataFrame and confirm the available columns. Step 5: Create Log_Feature1 by applying np.log() to Feature1. Step 6: Bin Feature2 into three equal-width categories using pd.cut(). Step 7: Label the categories as Low, Medium, and High. Step 8: Select the numerical columns required for correlation analysis. Step 9: Calculate and display the correlation matrix using corr(). Step 10: Count the number of observations in each Feature2 category. Step 11: Create a black-and-white scatter plot of Feature1 versus Feature2 grouped by Target. Step 12: Interpret the correlation, trend, and overlap visible in the output.			
Program	<pre>import pandas as pd import numpy as np import matplotlib.pyplot as plt # Predefined dictionary dataset data = { "Feature1": [5, 8, 12, 15, 18, 22, 25, 30, 35, 40, 45, 50, 55, 60, 70], "Feature2": [10, 18, 15, 25, 30, 28, 35, 40, 38, 45, 50, 55, 60, 65, 75], "Target": [0, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0, 1, 1] }</pre>			

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```
# Convert dictionary into a DataFrame
df = pd.DataFrame(data)
print("ORIGINAL DATAFRAME")
print(df)
print("DataFrame Shape:", df.shape)

# Create logarithmic feature
df["Log_Feature1"] = np.log(df["Feature1"])
print("\nDATAFRAME WITH LOG FEATURE")
print(df[["Feature1", "Log_Feature1"]].round(3))

# Bin Feature2 into three categories
df["Feature2_Category"] = pd.cut(
    df["Feature2"],
    bins=3,
    labels=["Low", "Medium", "High"]
)
print("\nFEATURE2 CATEGORIES")
print(df[["Feature2", "Feature2_Category"]])

# Calculate correlation matrix
numeric_columns = [
    "Feature1", "Feature2", "Target", "Log_Feature1"
]
correlation_matrix = df[numeric_columns].corr()
print("\nCORRELATION MATRIX")
print(correlation_matrix.round(3))

# Display category-wise counts
category_count = (
    df["Feature2_Category"]
    .value_counts()
    .sort_index()
)
print("\nFEATURE2 CATEGORY COUNT")
print(category_count)

# Create black-and-white scatter plot
class_zero = df["Target"] == 0
class_one = df["Target"] == 1

plt.figure(figsize=(8, 6))
plt.scatter(
    df.loc[class_zero, "Feature1"],
    df.loc[class_zero, "Feature2"],
    facecolors="white",
    edgecolors="black",
    marker="o",
```

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```
label="Target = 0"
)
plt.scatter(
    df.loc[class_one, "Feature1"],
    df.loc[class_one, "Feature2"],
    color="black",
    marker="s",
    label="Target = 1"
)
plt.title("Feature1 vs. Feature2 Grouped by Target")
plt.xlabel("Feature1")
plt.ylabel("Feature2")
plt.grid(True, linestyle="--", alpha=0.5)
plt.legend()
plt.tight_layout()
plt.show()

# Interpret visible patterns
feature_correlation = df["Feature1"].corr(df["Feature2"])
print("\nPATTERN INTERPRETATION")
print(
    "Correlation between Feature1 and Feature2:",
    round(feature_correlation, 3)
)
print(
    "Feature1 and Feature2 have a strong positive "
    "relationship. Target classes overlap, although "
    "Target = 1 appears more often at medium and high values."
)
```

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**Output
Snapshots
With
Explanation/
Observations**

Screenshot 1: Original DataFrame

ORIGINAL DATAFRAME

	Feature1	Feature2	Target
0	5	10	0
1	8	18	0
2	12	15	1
3	15	25	0
4	18	30	1
5	22	28	1
6	25	35	0
7	30	40	1
8	35	38	1
9	40	45	0
10	45	50	1
11	50	55	1
12	55	60	0
13	60	65	1
14	70	75	1

DataFrame Shape: (15, 3)

Figure 1.1 - Original DataFrame

Explanation/Observation: The dictionary was converted into a DataFrame containing 15 observations and the three required columns: Feature1, Feature2, and Target.

Screenshot 2: Logarithmic Feature

LOGARITHMIC FEATURE CREATED FROM FEATURE1

	Feature1	Log_Feature1
0	5	1.609
1	8	2.079
2	12	2.485
3	15	2.708
4	18	2.890
5	22	3.091
6	25	3.219
7	30	3.401
8	35	3.555
9	40	3.689
10	45	3.807
11	50	3.912
12	55	4.007
13	60	4.094
14	70	4.248

Figure 1.2 - Logarithmic Transformation of Feature1

Explanation/Observation: The natural logarithm of Feature1 was calculated using np.log(). The transformed values increase gradually and reduce the scale difference among larger values.

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Screenshot 3: Feature2 Binning

FEATURE2 BINNED INTO LOW, MEDIUM AND HIGH

	Feature2	Feature2_Category
0	10	Low
1	18	Low
2	15	Low
3	25	Low
4	30	Low
5	28	Low
6	35	Medium
7	40	Medium
8	38	Medium
9	45	Medium
10	50	Medium
11	55	High
12	60	High
13	65	High
14	75	High

Figure 1.3 - Feature2 Categorized as Low, Medium, and High

Explanation/Observation: The `pd.cut()` function divided Feature2 into three equal-width categories. The output contains 6 Low, 5 Medium, and 4 High observations.

Screenshot 4: Correlation Matrix

CORRELATION MATRIX

	Feature1	Feature2	Target	Log_Feature1
Feature1	1.000	0.990	0.336	0.943
Feature2	0.990	1.000	0.314	0.946
Target	0.336	0.314	1.000	0.400
Log_Feature1	0.943	0.946	0.400	1.000

Figure 1.4 - Correlation Matrix of Numerical Features

Explanation/Observation: Feature1 and Feature2 have a very strong positive correlation of approximately 0.990. Log_Feature1 also has strong positive relationships with both original features.

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Screenshot 5: Category Count and Pattern Interpretation

CATEGORY COUNT AND PATTERN INTERPRETATION

Feature2_Category

Low 6

Medium 5

High 4

Correlation between Feature1 and Feature2: 0.990

Interpretation: Feature1 and Feature2 show a strong positive relationship. Target classes overlap, although Target = 1 occurs more often at medium and high feature values.

Figure 1.5 - Category Counts and Console Interpretation

Explanation/Observation: The category distribution and calculated correlation confirm an increasing relationship between Feature1 and Feature2. Target classes overlap, so the features do not completely separate the two classes.

Screenshot 6: Scatter Plot

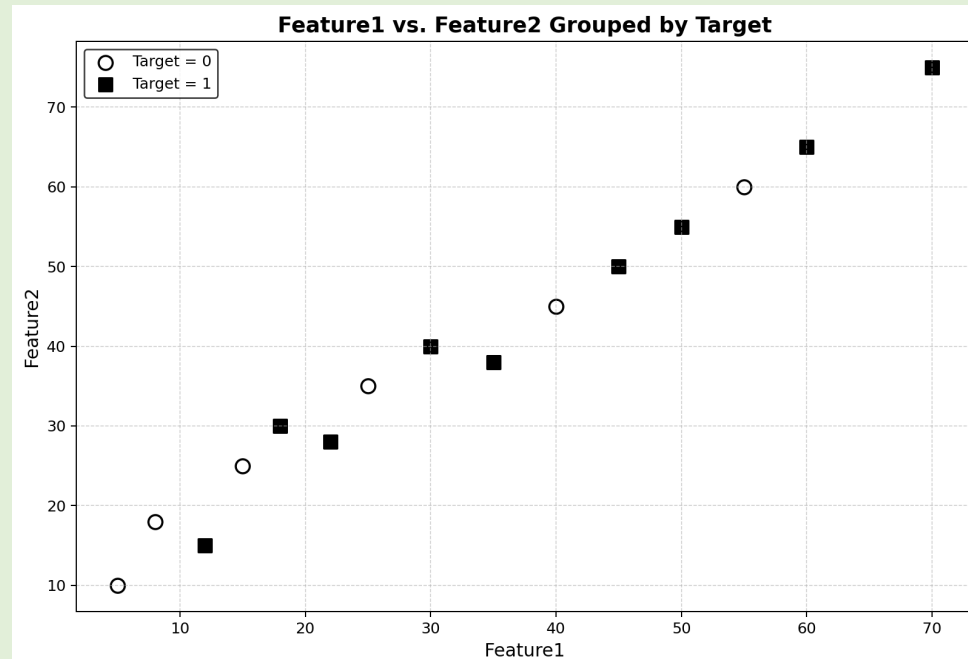


Figure 1.6 - Feature1 versus Feature2 Grouped by Target

Explanation/Observation: The points follow a clear upward pattern, showing that Feature2 generally increases as Feature1 increases. Black squares represent Target = 1, while white circles represent Target = 0. The classes overlap, but Target = 1 occurs more frequently at medium and high feature values.

Conclusions

The feature engineering and correlation analysis exercise was completed successfully using Pandas, NumPy, and Matplotlib. A DataFrame was created from a locally defined Python dictionary. A logarithmic feature was generated from Feature1, Feature2 was categorized into Low, Medium, and

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	High groups, and a correlation matrix was calculated for the numerical variables. The correlation value of approximately 0.990 showed a very strong positive relationship between Feature1 and Feature2. The black-and-white scatter plot confirmed the upward trend and showed that the Target classes overlap, although Target = 1 appears more frequently at medium and high feature values. The exercise provided practical understanding of feature transformation, binning, correlation analysis, visualization, and pattern interpretation.
References	1. Pandas documentation for DataFrame creation, binning, category counting, and correlation analysis. 2. NumPy documentation for logarithmic transformations. 3. Matplotlib documentation for scatter plots, legends, labels, and plot formatting. 4. Python Program (https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/WEEK%204/PRACTICE_PROGRAM.py) 5. Programming for Data Science (MACSE502) class lecture notes (https://drive.google.com/file/d/1PqsD1tC9aCgGFZo7QIJblBulplJg4UBs/view)